

Q-Learning Agent Report

Episode number	Average score
50	170.0
100	174.0
150	98.0
200	110.0
250	204.0
300	208.0
350	190.0
400	190.0
450	98.0
500	142.0

Average score of QLearningAgent vs StarterGhosts, tested every 50 training episodes over 5 games.

I trained the QLearningAgent for 500 episodes. Every 50 episodes I tested it for 5 games against StarterGhosts. The average scores stay between about 98 and 208. The values move up and down because each test uses only 5 games and the ghosts are random. This shows that the agent learns some useful behaviour, but it does not become very strong.

50-game evaluation screenshot (RunQLearningGames output)

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Q learning evaluation vs StarterGhosts
Game    Score
0       80
1       90
2      210
3       90
4       90
5       90
6       80
7       90
8       80
9       80
10      80
11      90
12     100
13      80
14      80
15     190
16      90
17      80
18     150
19      80
20     130
21     120
22      90
23      90
24      80
25      80
26      80
27      90
28      90
29      90
30      80
31      90
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30 80
31 90
32 90
33 90
34 80
35 80
36 80
37 130
38 80
39 90
40 80
41 90
42 130
43 130
44 90
45 290
46 100
47 100
48 150
49 80

Average score: 101.4
Wins: 0 Losses: 50
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After training I ran 50 games against StarterGhosts. The agent reached an average score of 101.4 and did not win any game. This means the learned policy is a bit better than random play, but it is not strong enough to clear the maze.